

SECTION C

10 (short)

- (a) For metallic conductors explain why it is assumed that any net charge is distributed evenly on the surface of the conductor. [2]
- (b) Two point charges of $5\ \mu\text{C}$ and $-7\ \mu\text{C}$ are placed in free space at positions $(1,0)$ and $(1,1)$ respectively on a two dimensional (x,y) plane where coordinates are measured in centimetres.
- (i) Find the electric field due to the two point charges at coordinate $(-1,-1)$, expressing the field in terms of magnitude and direction. [6]
- (ii) If a charge of $10\ \mu\text{C}$ is now inserted at coordinate $(-1, -1)$, determine the magnitude of the force acting upon it. [2]

11 (short)

- (a) Define the self inductance of a circuit. [2]
- (b) Determine the self inductance of a solenoid with n turns per unit length, length l and cross-sectional area A . Assume that the length l is much greater than the diameter of the solenoid cross-section. [3]
- (c) Figure 6 shows a simple circuit containing an inductor of value L and resistor R connected through a switch at position K_1 to a DC source. The current flowing through the circuit in the steady state is I_0 . The switch is then moved to position K_2 . Derive an expression for the current in the circuit as a function of time. [5]

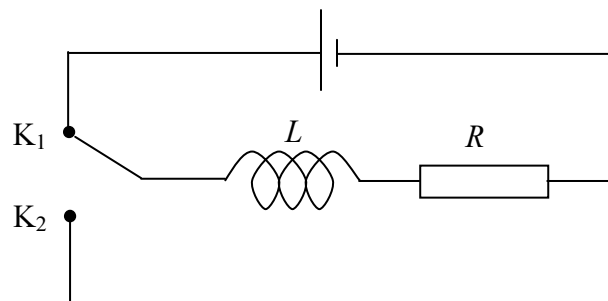


Fig. 6